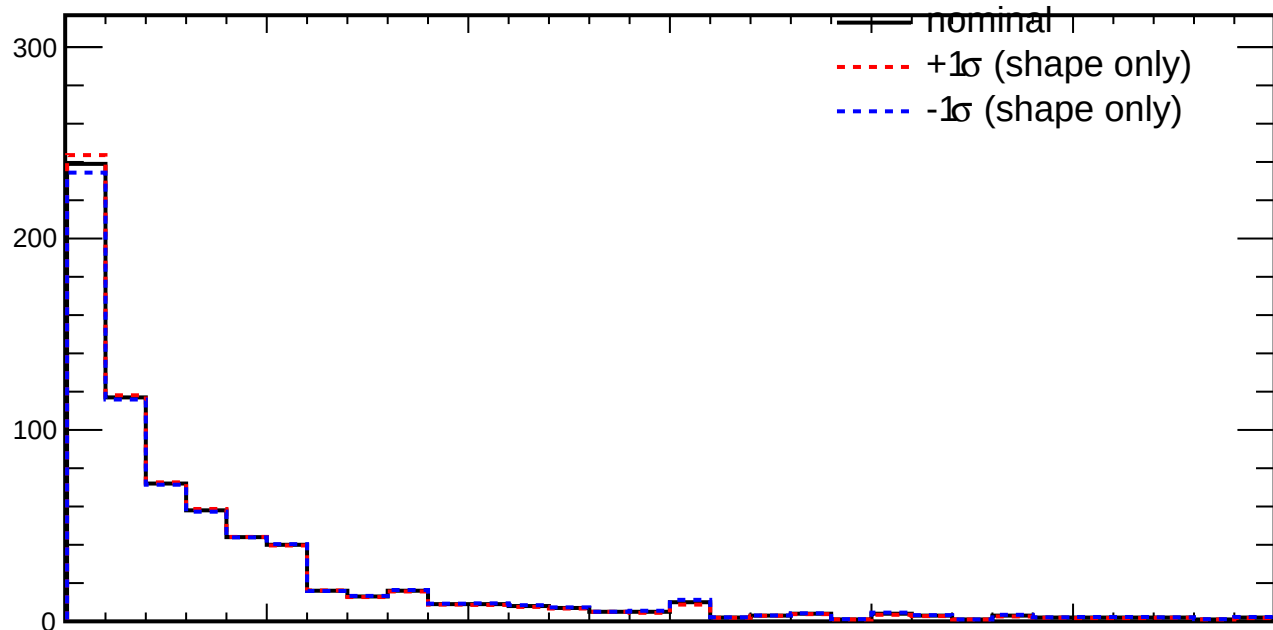
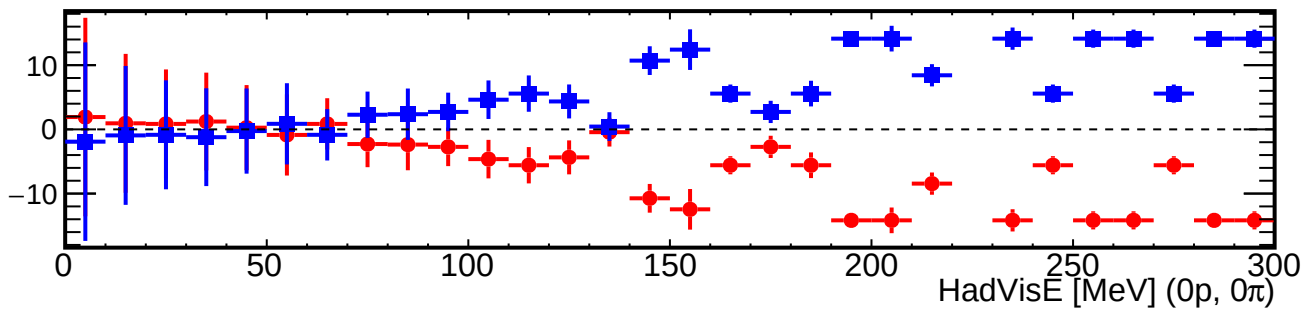


FHC [QE]: ZNormCCQE (shape only) $\chi^2_{\text{shape},+1\sigma}=0.77$ $\chi^2_{\text{shape},-1\sigma}=0.76$ $\Delta\langle x\rangle_{+1\sigma}=-1.5911$ $\Delta\langle x\rangle_{-1\sigma}=+1.5868$
 $\chi^2_{\text{raw},+1\sigma}=20.09$ $\chi^2_{\text{raw},-1\sigma}=11.30$ rate: +16.51% / -12.38% $\Delta\sigma^2$: -204.1582 / +198.5656

Events

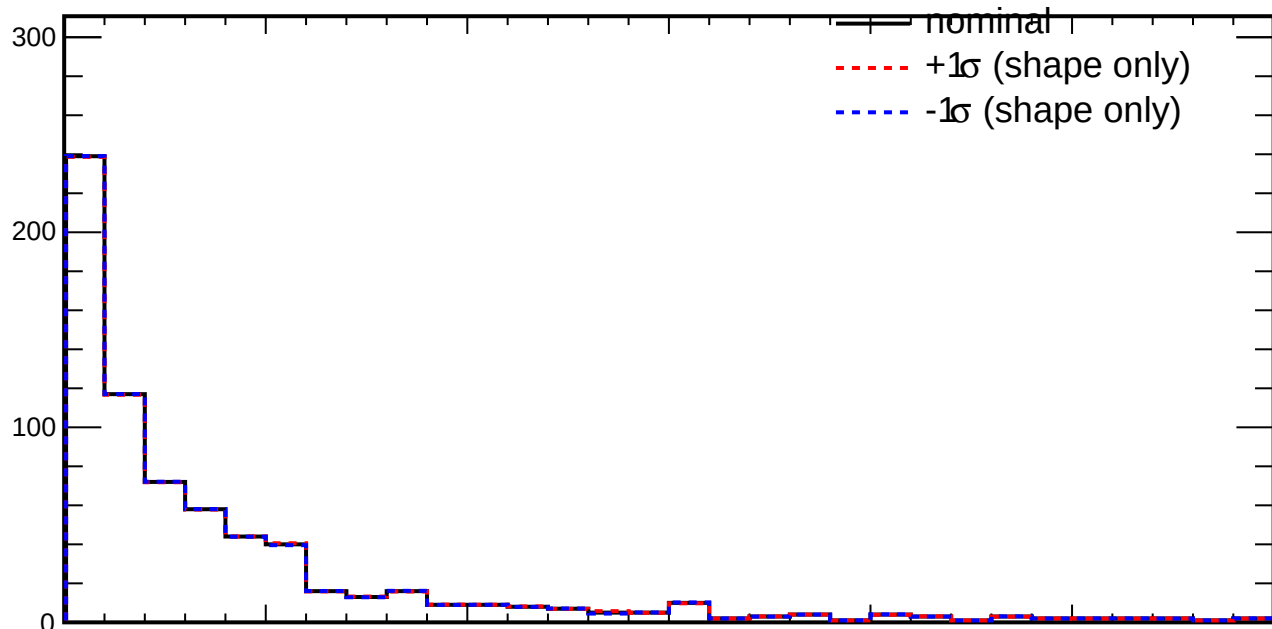


% diff from nominal

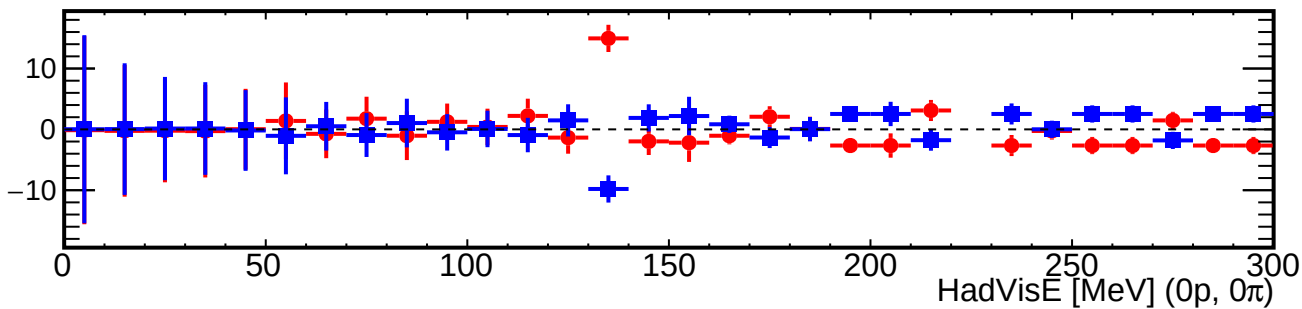


FHC [QE]: RPAshapeenh2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.16$ $\chi^2_{\text{shape},-1\sigma}=0.08$ $\Delta\langle x \rangle_{+1\sigma}=+0.0352$ $\Delta\langle x \rangle_{-1\sigma}=+0.0558$
 $\chi^2_{\text{raw},+1\sigma}=0.69$ $\chi^2_{\text{raw},-1\sigma}=0.50$ rate: +2.73% / -2.47% $\Delta\sigma^2$: -12.4082 / +17.0853

Events

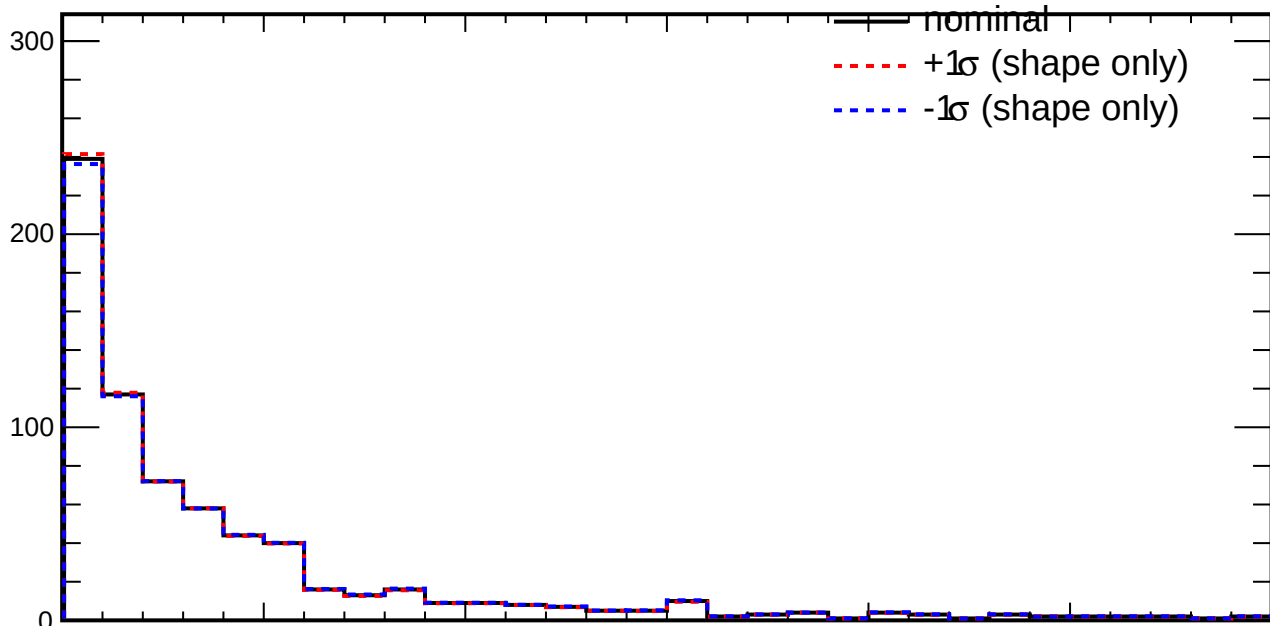


% diff from nominal

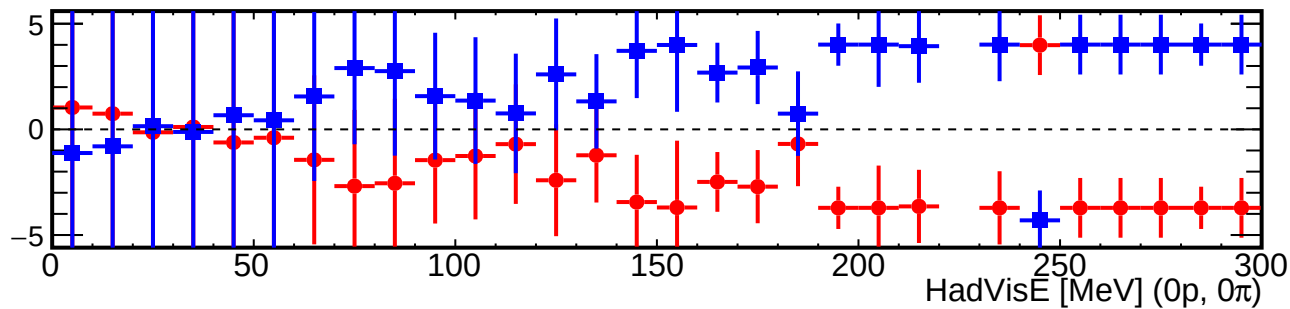


FHC [QE]: RPAShapesupp2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.12$ $\chi^2_{\text{shape},-1\sigma}=0.14$ $\Delta\langle x\rangle_{+1\sigma}=-0.5737$ $\Delta\langle x\rangle_{-1\sigma}=+0.6197$
 $\chi^2_{\text{raw},+1\sigma}=1.17$ $\chi^2_{\text{raw},-1\sigma}=1.17$ rate: +3.86% / -3.86% $\Delta\sigma^2$: -54.9589 / +58.6287

Events

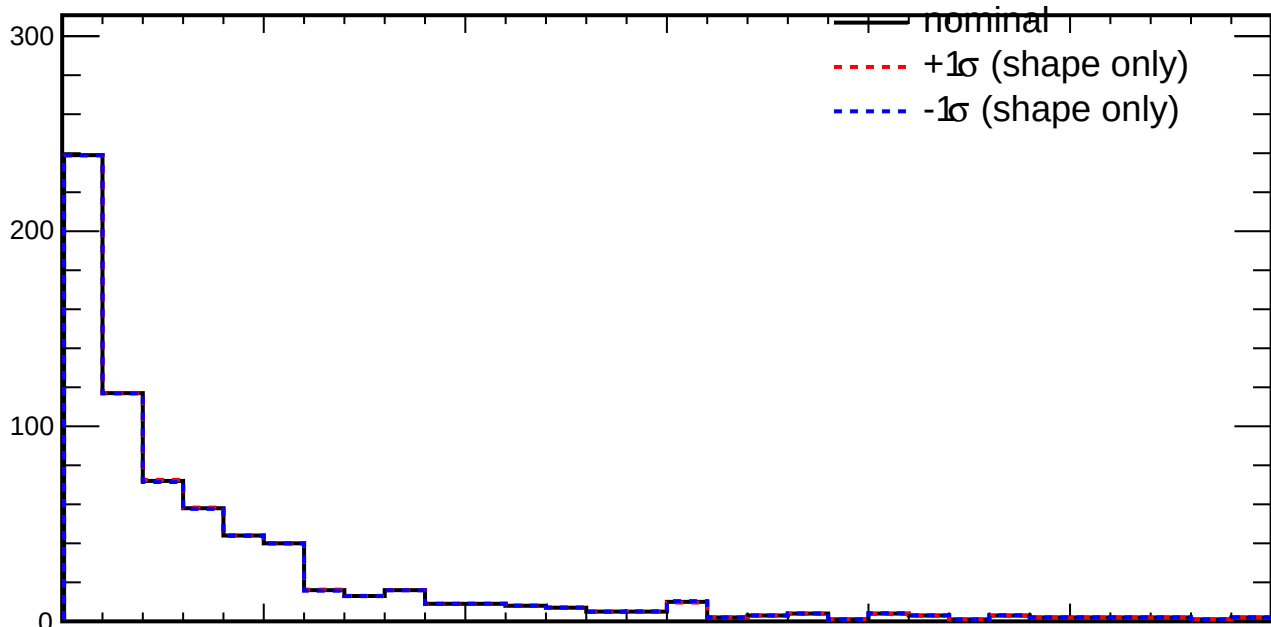


% diff from nominal

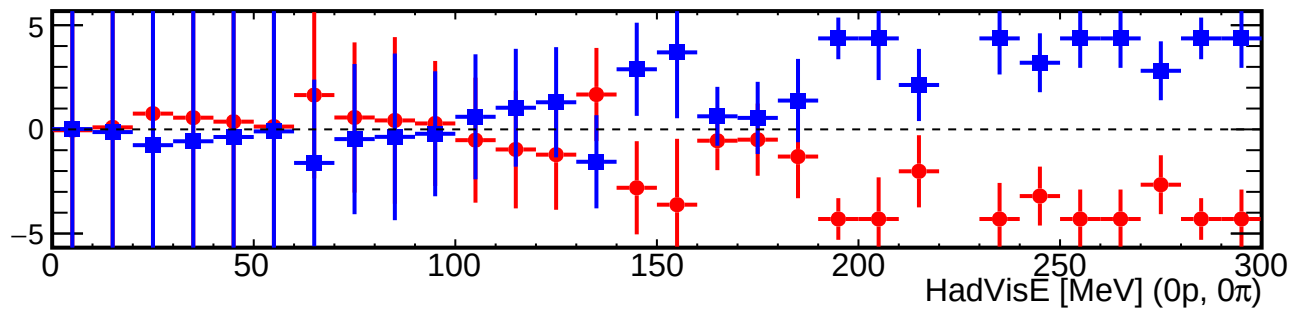


FHC [QE]: ZExpAxialFFSyst2020_EV1 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.07$ $\chi^2_{\text{shape},-1\sigma}=0.07$ $\Delta\langle x \rangle_{+1\sigma}=-0.3373$ $\Delta\langle x \rangle_{-1\sigma}=+0.3554$
 $\chi^2_{\text{raw},+1\sigma}=1.48$ $\chi^2_{\text{raw},-1\sigma}=1.29$ rate: +4.50% / -4.18% $\Delta\sigma^2$: -60.9373 / +62.2522

Events

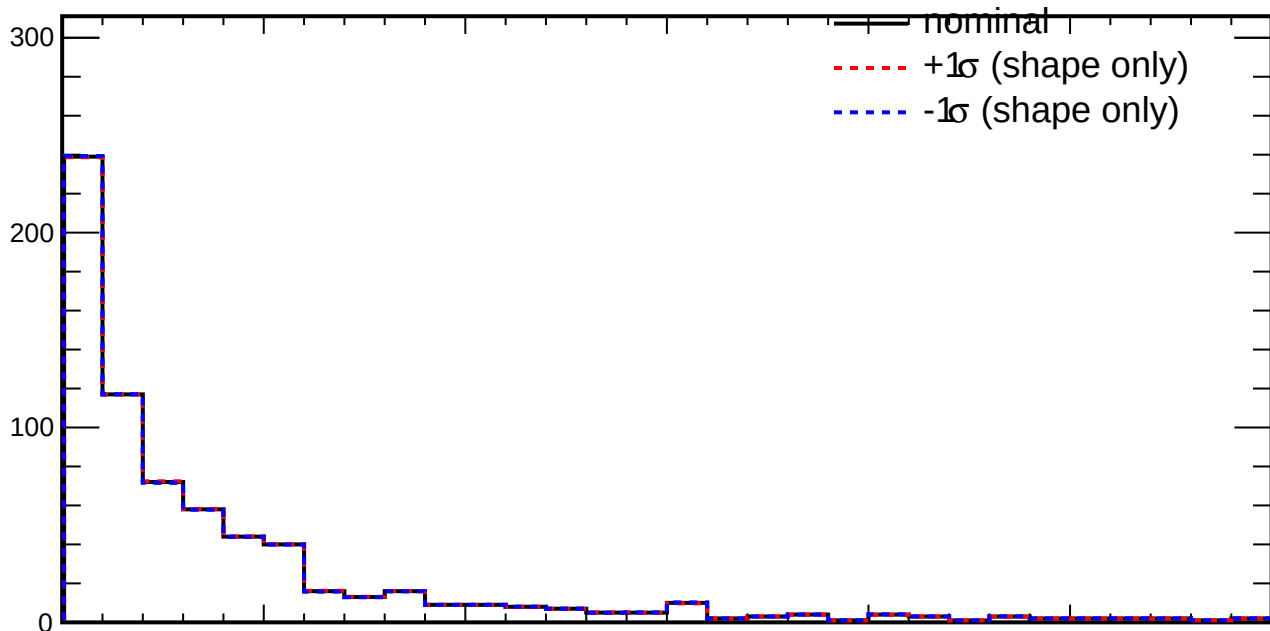


% diff from nominal

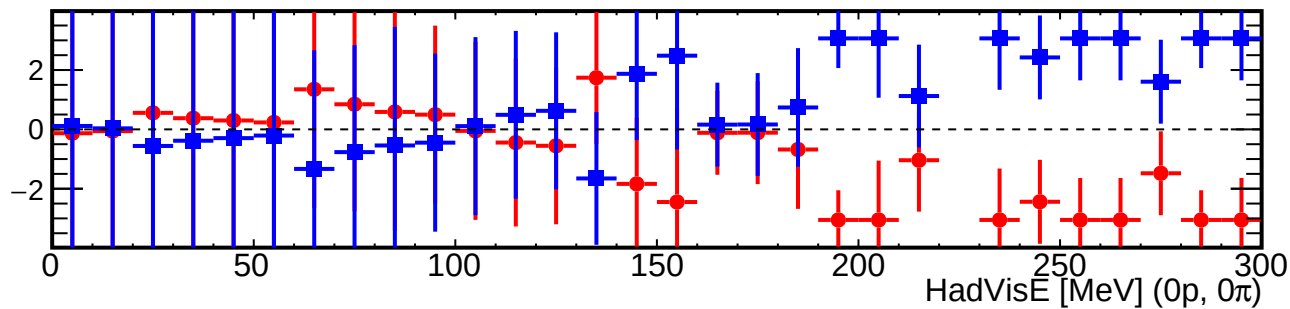


FHC [QE]: ZExpAxialFFSyst2020_EV2 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.03$ $\chi^2_{\text{shape},-1\sigma}=0.03$ $\Delta\langle x\rangle_{+1\sigma}=-0.1888$ $\Delta\langle x\rangle_{-1\sigma}=+0.1993$
 $\chi^2_{\text{raw},+1\sigma}=0.73$ $\chi^2_{\text{raw},-1\sigma}=0.65$ rate: +3.15% / -2.97% $\Delta\sigma^2$: -40.5109 / +41.2264

Events

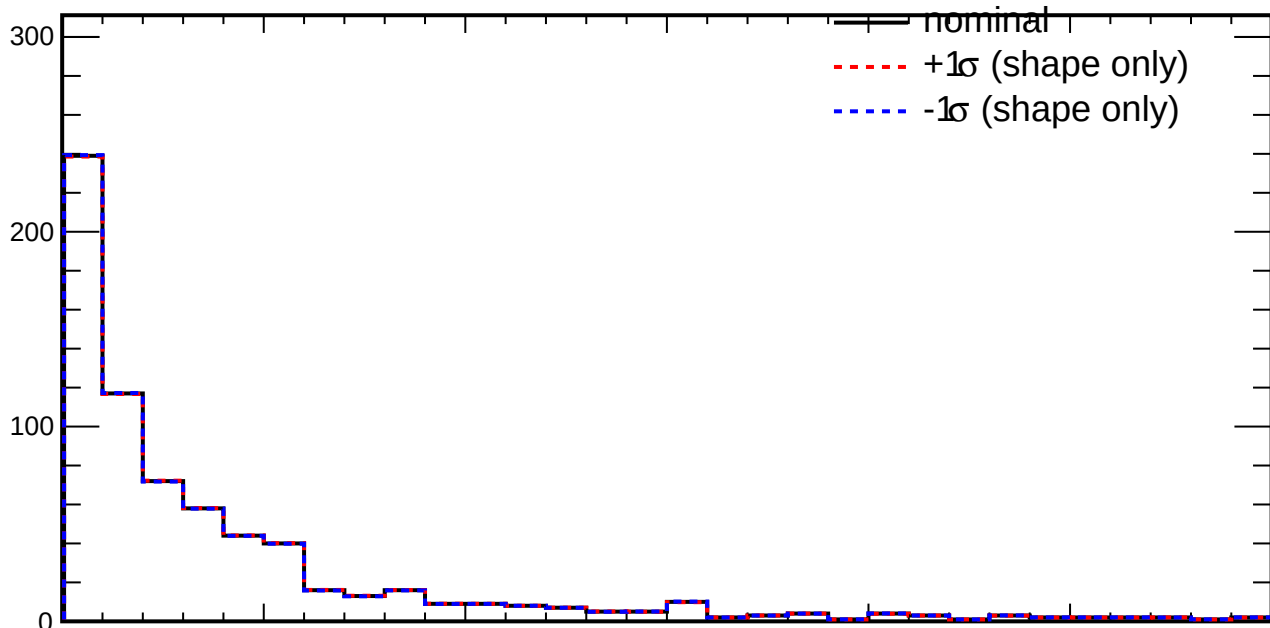


% diff from nominal

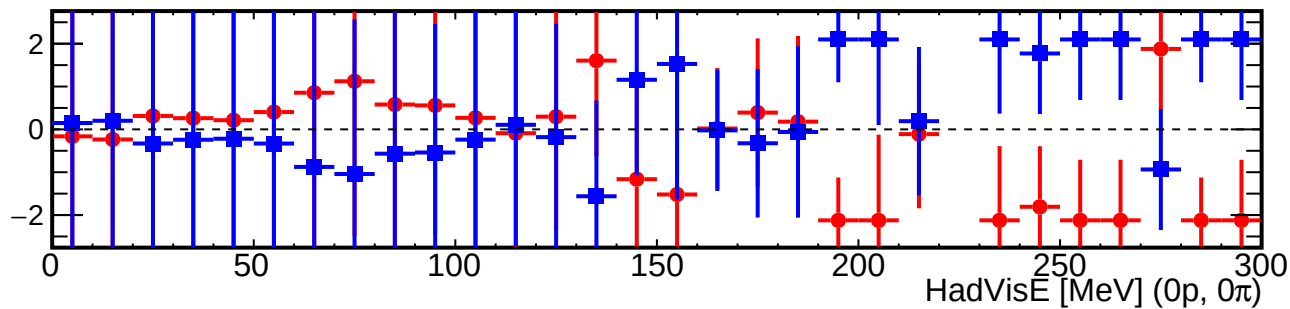


FHC [QE]: ZExpAxialFFSyst2020_EV3 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.02$ $\chi^2_{\text{shape},-1\sigma}=0.02$ $\Delta\langle x \rangle_{+1\sigma}=-0.0563$ $\Delta\langle x \rangle_{-1\sigma}=+0.0697$
 $\chi^2_{\text{raw},+1\sigma}=0.35$ $\chi^2_{\text{raw},-1\sigma}=0.31$ rate: +2.17% / -2.06% $\Delta\sigma^2$: -20.3960 / +22.0047

Events

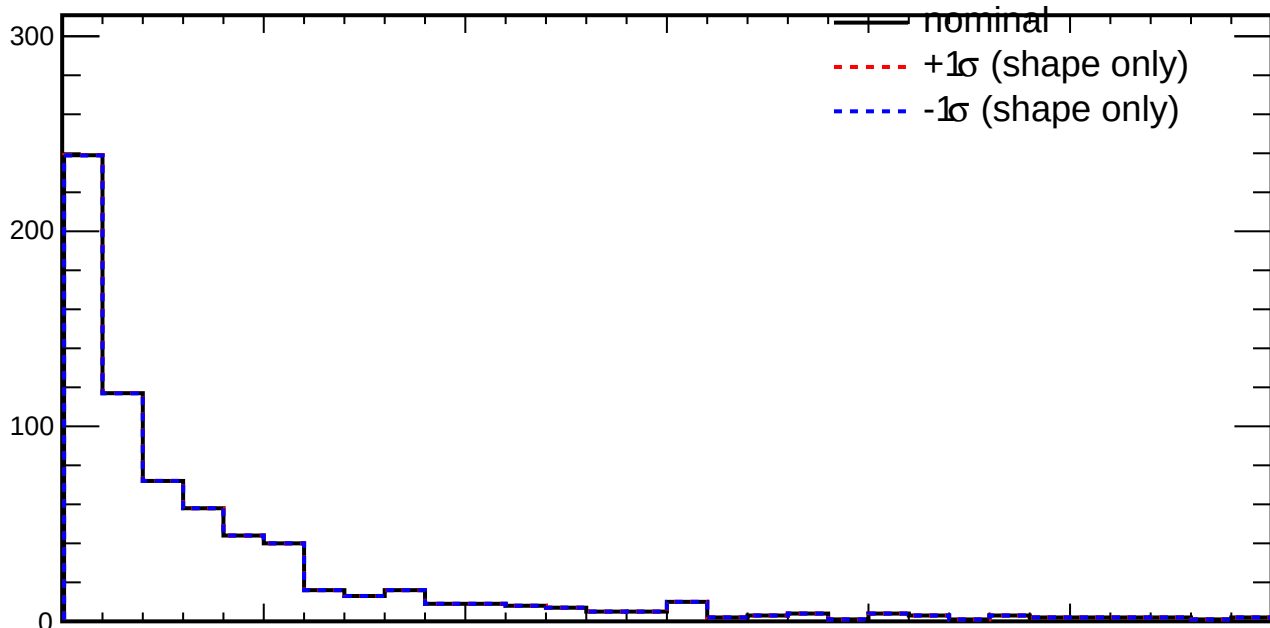


% diff from nominal

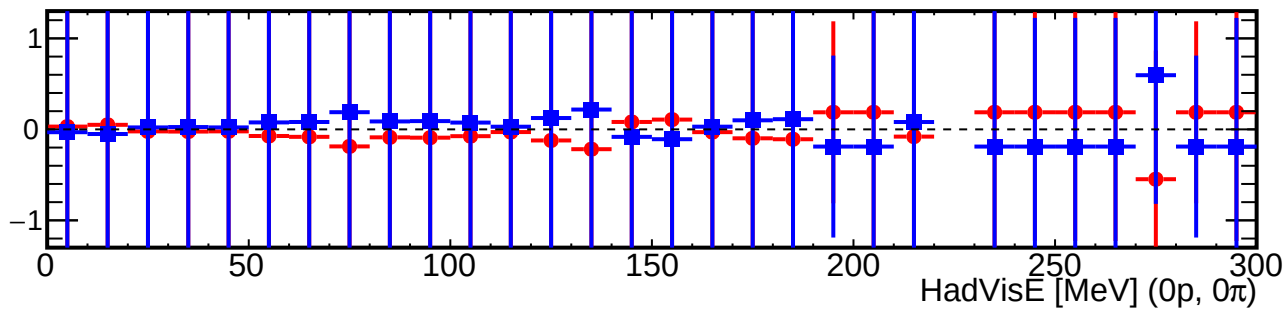


FHC [QE]: ZExpAxialFFSyst2020_EV4 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0069$ $\Delta\langle x\rangle_{-1\sigma}=+0.0073$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: -0.19% / +0.19% $\Delta\sigma^2$: +0.7386 / -0.6729

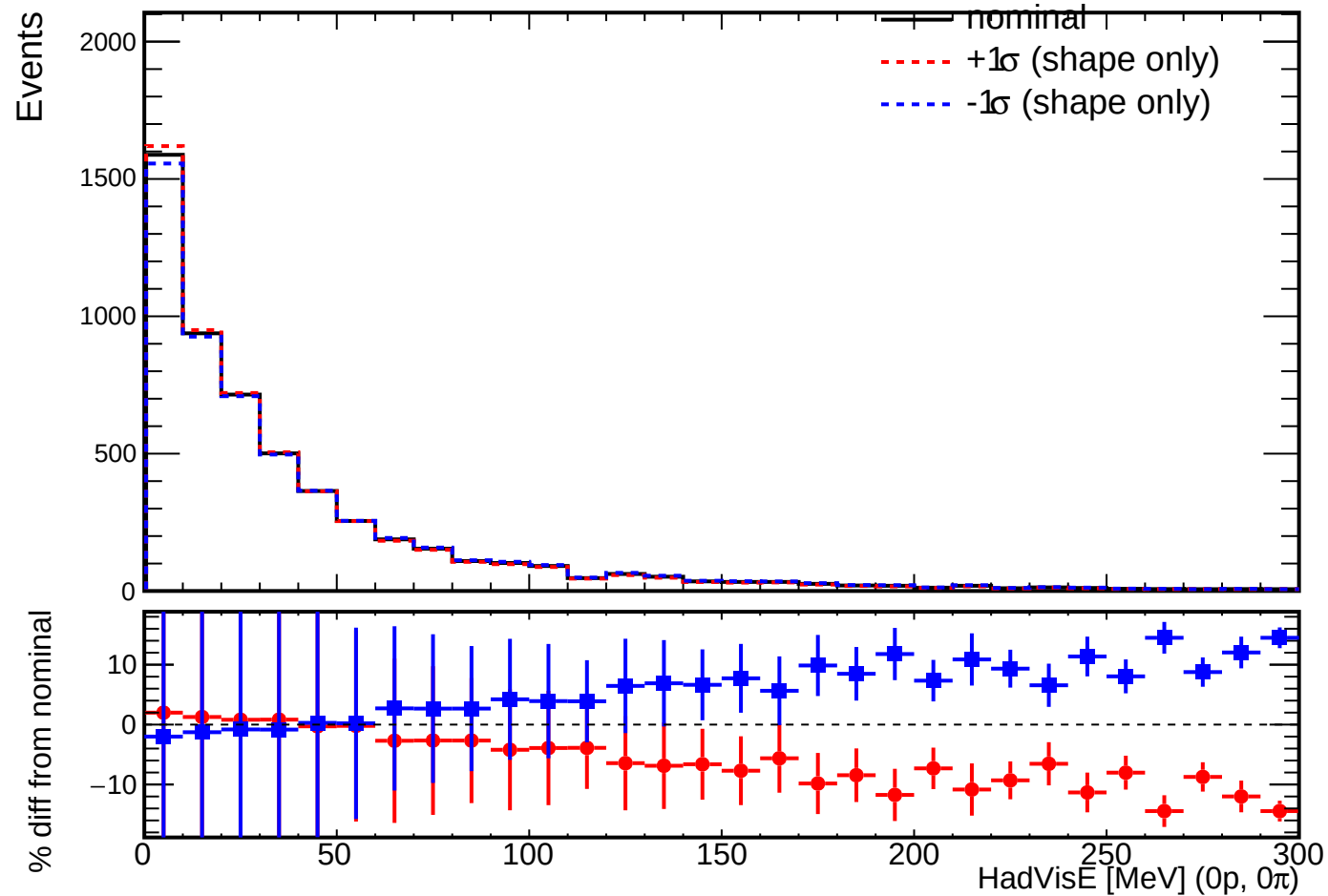
Events



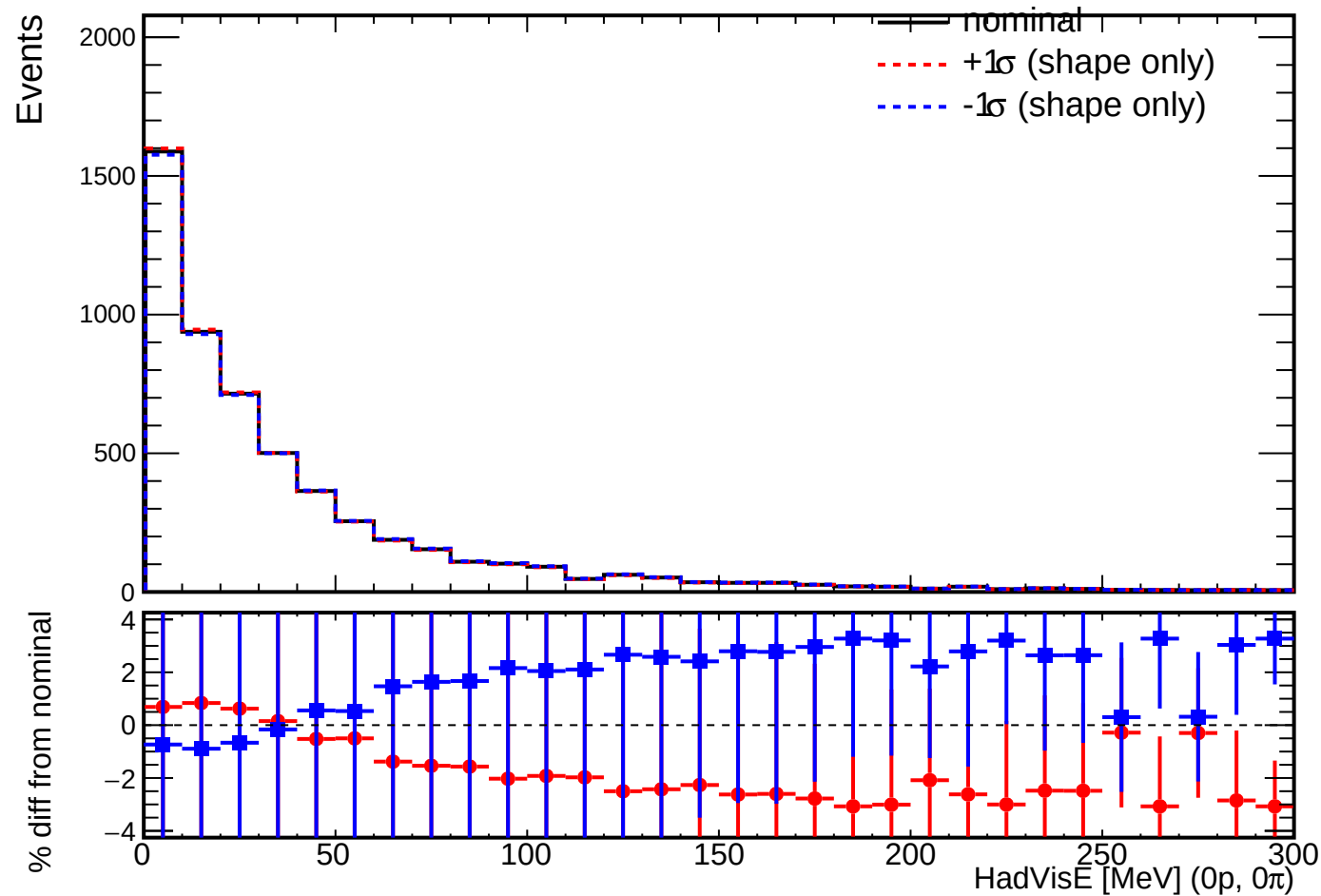
% diff from nominal



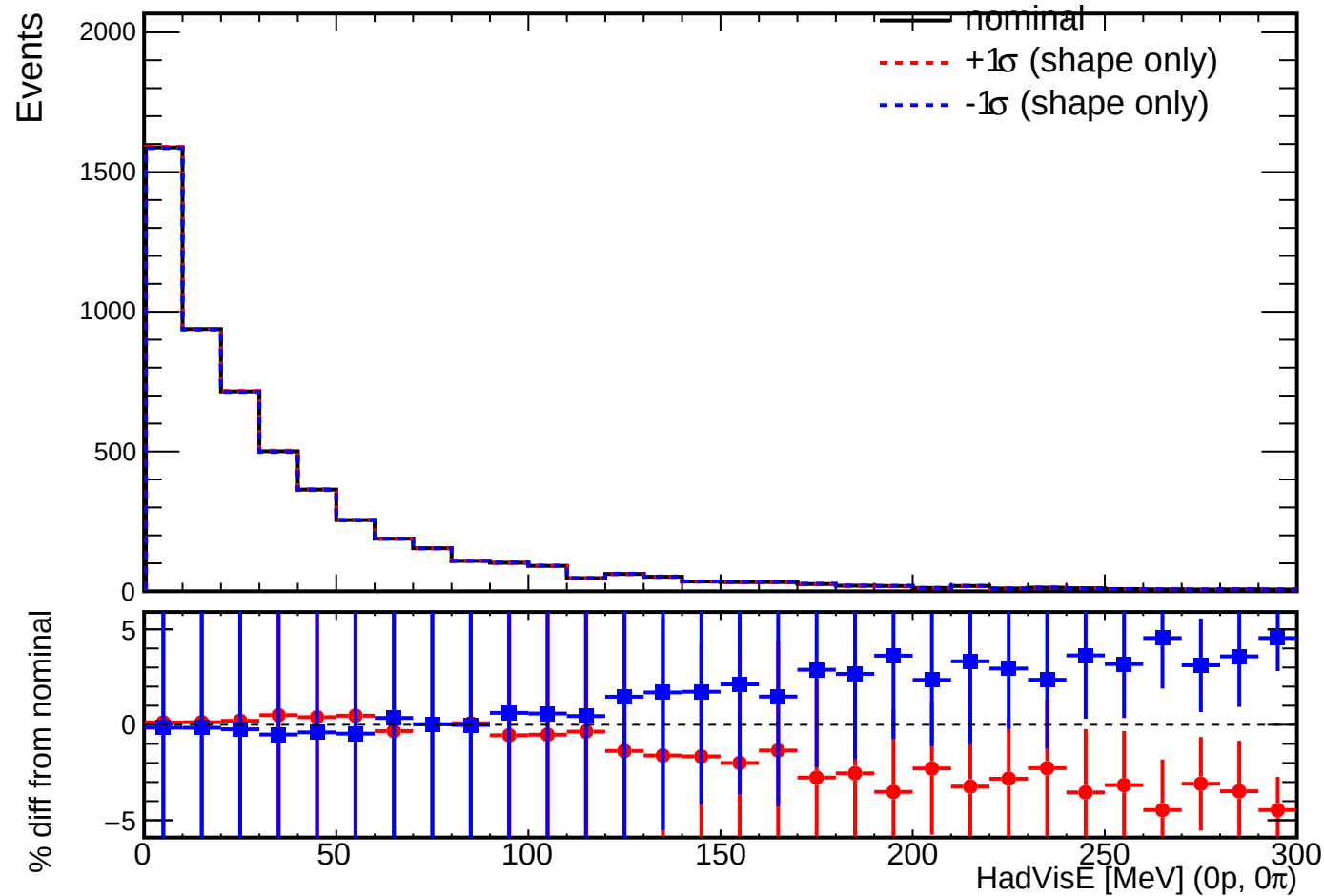
RHC [QE]: ZNormCCQE (shape only) $\chi^2_{\text{shape},+1\sigma}=4.17$ $\chi^2_{\text{shape},-1\sigma}=4.20$ $\Delta\langle x \rangle_{+1\sigma}=-1.2557$ $\Delta\langle x \rangle_{-1\sigma}=+1.2602$
 $\chi^2_{\text{raw},+1\sigma}=160.27$ $\chi^2_{\text{raw},-1\sigma}=90.15$ rate: +16.88% / -12.66% $\Delta\sigma^2$: -130.0408 / +127.3365



RHC [QE]: RPAShapesupp2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.62$ $\chi^2_{\text{shape},-1\sigma}=0.71$ $\Delta\langle x\rangle_{+1\sigma}=-0.4513$ $\Delta\langle x\rangle_{-1\sigma}=+0.4808$
 $\chi^2_{\text{raw},+1\sigma}=6.12$ $\chi^2_{\text{raw},-1\sigma}=6.12$ rate: +3.17% / -3.17% $\Delta\sigma^2$: -36.4029 / +38.3387

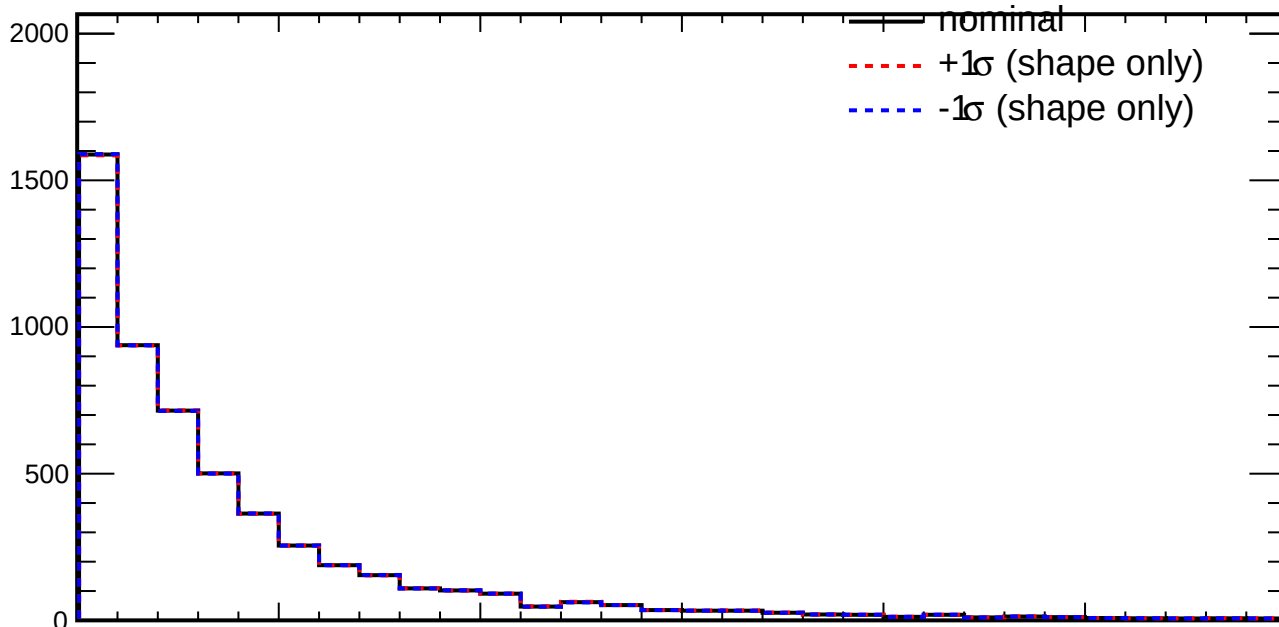


RHC [QE]: ZExpAxialFFSyst2020_EV1 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.25$ $\chi^2_{\text{shape},-1\sigma}=0.27$ $\Delta\langle x\rangle_{+1\sigma}=-0.2611$ $\Delta\langle x\rangle_{-1\sigma}=+0.2777$
 $\chi^2_{\text{raw},+1\sigma}=12.12$ $\chi^2_{\text{raw},-1\sigma}=10.50$ rate: +4.67% / -4.35% $\Delta\sigma^2$: -37.7185 / +38.9594

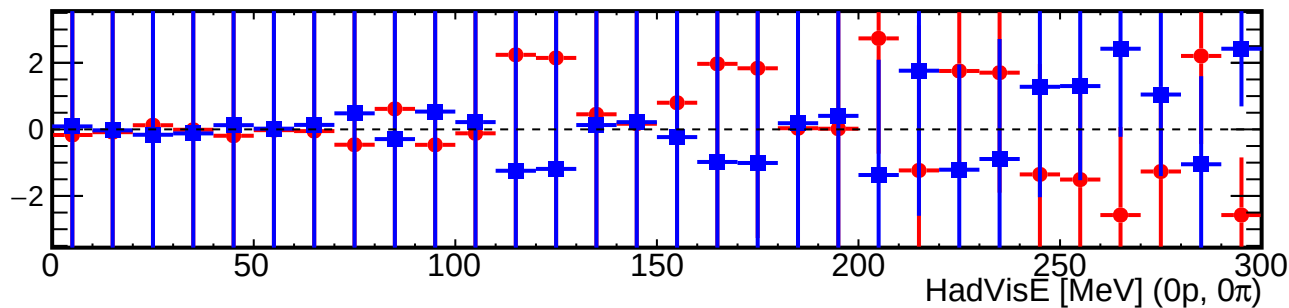


RHC [QE]: RPAShapeenh2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.13$ $\chi^2_{\text{shape},-1\sigma}=0.06$ $\Delta\langle x\rangle_{+1\sigma}=+0.0841$ $\Delta\langle x\rangle_{-1\sigma}=-0.0049$
 $\chi^2_{\text{raw},+1\sigma}=3.93$ $\chi^2_{\text{raw},-1\sigma}=3.09$ rate: +2.64% / -2.37% $\Delta\sigma^2$: +5.9152 / +1.6785

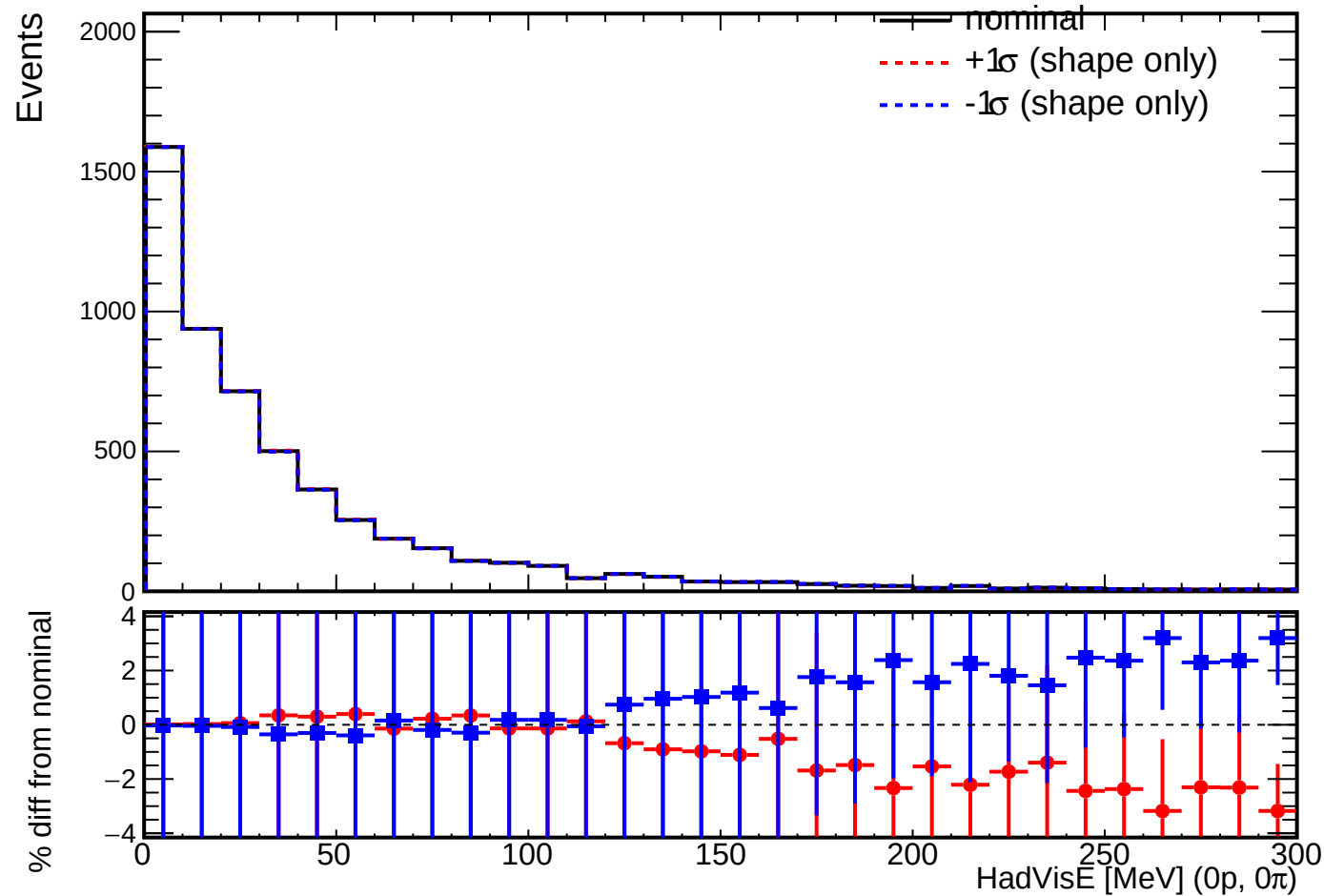
Events



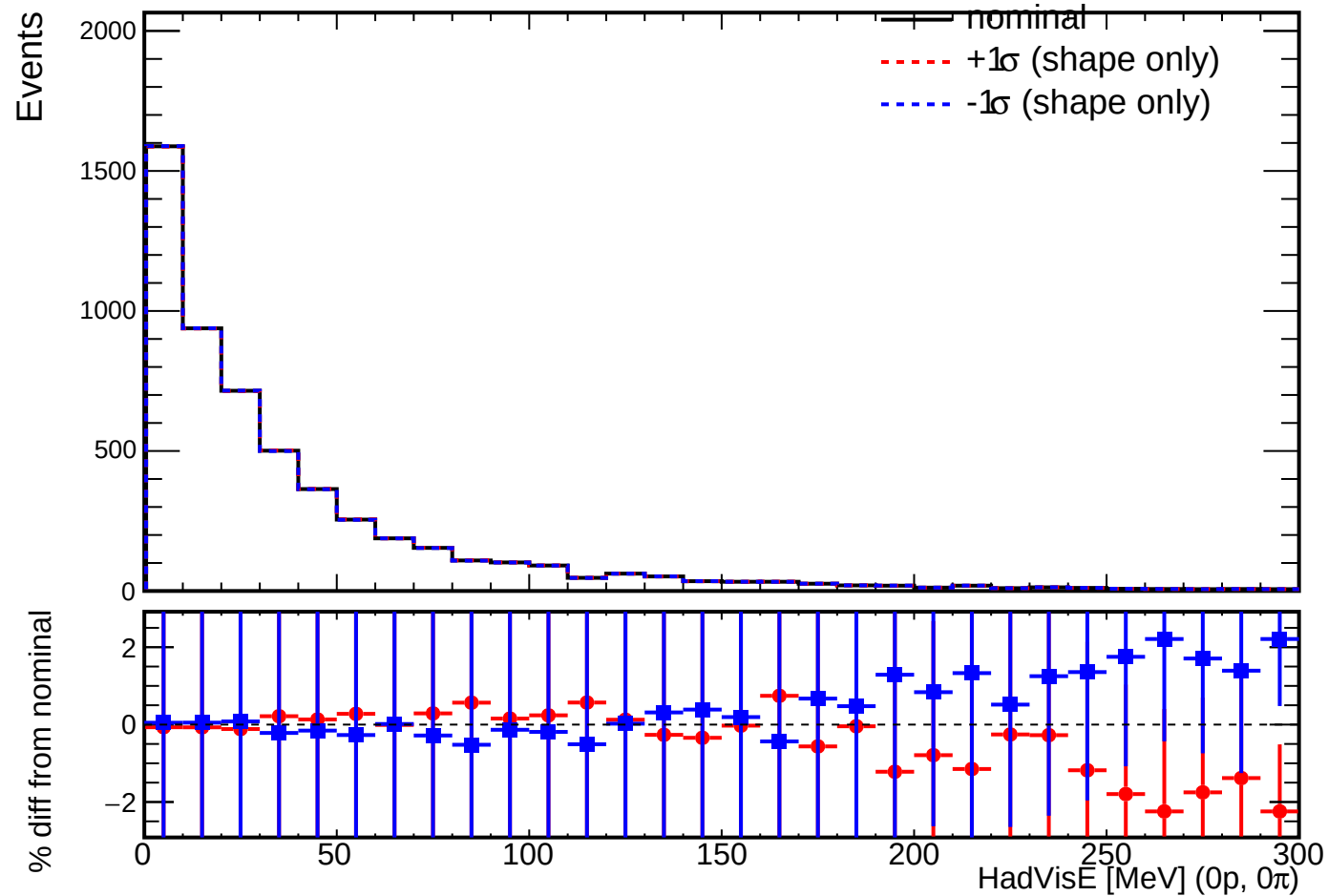
% diff from nominal



RHC [QE]: ZExpAxialFFSyst2020_EV2 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.10$ $\chi^2_{\text{shape},-1\sigma}=0.11$ $\Delta\langle x \rangle_{+1\sigma}=-0.1403$ $\Delta\langle x \rangle_{-1\sigma}=+0.1501$
 $\chi^2_{\text{raw},+1\sigma}=5.95$ $\chi^2_{\text{raw},-1\sigma}=5.31$ rate: +3.28% / -3.10% $\Delta\sigma^2$: -23.8927 / +24.6358



RHC [QE]: ZExpAxialFFSyst2020_EV3 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.04$ $\chi^2_{\text{shape},-1\sigma}=0.04$ $\Delta\langle x \rangle_{+1\sigma}=-0.0202$ $\Delta\langle x \rangle_{-1\sigma}=+0.0417$
 $\chi^2_{\text{raw},+1\sigma}=2.90$ $\chi^2_{\text{raw},-1\sigma}=2.58$ rate: +2.29% / -2.16% $\Delta\sigma^2$: -9.4846 / +12.0007



RHC [QE]: ZExpAxialFFSyst2020_EV4 (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x \rangle_{+1\sigma}=-0.0090$ $\Delta\langle x \rangle_{-1\sigma}=+0.0102$
 $\chi^2_{\text{raw},+1\sigma}=0.02$ $\chi^2_{\text{raw},-1\sigma}=0.02$ rate: -0.20% / +0.21% $\Delta\sigma^2: +0.0308$ / +0.1368

